

PLANT SAFETY MANUAL - SECTION 7: ROTATING EQUIPMENT  
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[FICTIONAL/DEMO DOCUMENT]

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7.1 ROTATING EQUIPMENT HAZARDS  
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Key hazards associated with pumps and rotating equipment:

- a) Entanglement in rotating parts
- b) High pressure fluid release
- c) High temperature surfaces
- d) Electrical hazards
- e) Hydrocarbon release and fire risk
- f) Noise-induced hearing loss

7.2 EQUIPMENT OPERATING LIMITS  
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Temperature Alarms:

- High temperature alarm (TA-High): 85 degrees Celsius
- High-high temperature alarm (TA-HH): 95 degrees Celsius
- Automatic trip on high-high alarm

Vibration Alarms:

- High vibration alarm (VA-High): 7.0 mm/s
- High-high vibration alarm (VA-HH): 11.0 mm/s
- Automatic trip on high-high alarm

Pressure Monitoring:

- Suction pressure low (PA-Low): 0.5 bar
- Discharge pressure high (PA-High): 20 bar

7.3 EMERGENCY PROCEDURES  
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Pump Failure Emergency Response:

1. Immediately activate standby pump (if available)
2. Notify control room operator
3. Isolate failed pump using emergency isolation valves
4. Initiate shutdown procedure per SOP-MECH-042
5. Assess for any fluid release and contain if necessary
6. Notify section head and safety officer
7. Complete incident report within 2 hours

7.4 PERMIT-TO-WORK

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Required for all maintenance activities:

- Hot Work Permit for welding, grinding, cutting
- Confined Space Entry Permit for enclosed vessels
- Electrical Isolation Permit for electrical work
- General Work Permit for mechanical maintenance

[THIS IS FICTIONAL DEMO DATA - FOR DEMONSTRATION ONLY]